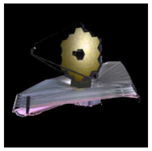

7605 - Disclosing the nature of GN-z11 with MIRI rest-frame near-IR imaging

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
MIRI - GNz11				
	1	GNz11 - MIRI F1000 W	MIRI Imaging	(1) GNz11
	2	GNz11 - MIRI F1280 W	MIRI Imaging	(1) GNz11

ABSTRACT

The galaxy GN-z11 at z=10.6 is the brightest known galaxy before the main Epoch of Reionization (EoR). GN-z11 belongs to the class of luminous galaxies in the early Universe detected by JWST in numberswell above predicted by the models. The nature of this galaxy as an extreme young and compact starburst or as a massive super-Eddington accreting black hole is under debate based on existing NIRCam and NIRSpec data covering a limited rest-frame spectral range (<0.4 microns). This program proposes MIRI F1000W and F1280W imaging of GN-z11, securing for the first time the detection of the rest-frame 0.86 and 1.1 micron continuum emission in a pre-EoR galaxy. These new data, together with archival MIRI F560W and F770W, will be able to distinguish between the near-IR emission of an accreting massive black hole (i.e. AGN) and that of different stellar populations. The proposed imaging will provide the key measurements to establish the star formation history and the nature of the dominant ionizing source, an extreme compact young starburst or an early AGN. This program, together with existing public JWST data, gives a unique opportunity to identify the nature of GN-z11, constraining the history of the early stellar mass buildup and the BH growth in a galaxy well before the main Epoch of Reionization, when the Universe was just 250-400 Myr old.

OBSERVING DESCRIPTION

We propose to conduct deep MIRI imaging of GN-z11 using the F1000W and F1280W filters. The primary aim of these observations is to detect, for the first time, the rest-frame near-IR continuum of a galaxy at the Cosmic Dawn (z>10). The near-IR emission of GN-z11 will be key to determine the star formation history and the nature of the dominant ionizing source in this galaxy, when the Universe was just ~300 Myr old.

We request the images to be taken in the FASTR1 readout mode with ~100 groups per integration and 9 and 13 integrations, obtaining about 25000 and 60000 seconds for the F1000W and F1280W filters, respectively. These observations are conducted using 10-16 dithers with the CYCLING large pattern to optimize the sampling of the MIRI PSF and the correction for residual cosmic rays. We constrain the V3 angle to optimize the overlap with ancillary MIRI F560W+F770W (ID 1264) and NIRCam (ID 1181) deep imaging of this field.

We splitted the proposal in 2 observations and we request group observations to occur within 4 days to minimize the impact of possibles changes in the background emission and V3 orientation.

Note*: We drop the PA constraints so that the observing window widens, providing more opportunities to be scheduled in 2026.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	GNz11	RA: 12 36 25.4600 (189.1060833d) Dec: +62 14 31.40 (62.24206d) Equinox: J2000		

Comments:
Category=Galaxy
Description=[Emission line galaxies, High-redshift galaxies, Primordial galaxies]
Extended=NO

Observation

Proposal 7605, Observation 1: GNz11 - MIRI F1000W

Thu Feb 05 15:00:11 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	GNz11	RA: 12 36 25.4600 (189.1060833d) Dec: +62 14 31.40 (62.24206d) Equinox: J2000		

Comments:
Category=Galaxy
Description=[Emission line galaxies, High-redshift galaxies, Primordial galaxies]
Extended=NO

Template

Subarray
FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	CYCLING	1	10		1	1			LARGE

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F1000W	FASTR1	100	9	1	Dither 1	10	90	25197.363	

Special Requirements

Group Observations 1, 2 within 4 Days

Observation

Proposal 7605, Observation 2: GNz11 - MIRI F1280W

Thu Feb 05 15:00:11 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	GNz11	RA: 12 36 25.4600 (189.1060833d) Dec: +62 14 31.40 (62.24206d) Equinox: J2000		

Comments:
Category=Galaxy
Description=[Emission line galaxies, High-redshift galaxies, Primordial galaxies]
Extended=NO

Template

Subarray
FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	CYCLING	1	16						LARGE

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F1280W	FASTR1	103	13	1	Dither 1	16	208	59985.265	

Special Requirements

Group Observations 1, 2 within 4 Days